

EXPERIMENT 1 - INPUT

Test Case 1: User Input

Enter number of nodes: 5

Enter nodes in level order: 1 2 3 4 5

Test Case 2: User Input

Enter number of nodes: 3

Enter nodes in level order: 10 20 30

EXPERIMENT 1 - OUTPUT

Test Case 1: Execution Result

Inorder Traversal: 4 2 5 1 3

Test Case 2: Execution Result

Inorder Traversal: 20 10 30

EXPERIMENT 2 - INPUT

Test Case 1: User Input

Enter level-order nodes: 1 2 3 4 5

Test Case 2: User Input

Enter level-order nodes: 10 20 30

EXPERIMENT 2 - OUTPUT

Test Case 1: Execution Result

```
Preorder Traversal:  
1 2 4 5 3
```

Test Case 2: Execution Result

```
Preorder Traversal:  
10 20 30
```

EXPERIMENT 3 - INPUT

Test Case 1: User Input

Enter number of nodes: 5

Enter nodes in level order: 1 2 3 4 5

Test Case 2: User Input

Enter number of nodes: 3

Enter nodes in level order: 10 20 30

EXPERIMENT 3 - OUTPUT

Test Case 1: Execution Result

Postorder Traversal: 4 5 2 3 1

Test Case 2: Execution Result

Postorder Traversal: 20 30 10

EXPERIMENT 4 - INPUT

Test Case 1: User Input

Enter input nodes: 5 2 8 1 3

Test Case 2: User Input

Enter input nodes: 10 20 30 40 50 60 70

EXPERIMENT 4 - OUTPUT

Test Case 1: Execution Result

Node Count : 5

Test Case 2: Execution Result

Node Count : 7

EXPERIMENT 5 - INPUT

Test Case 1: User Input

Enter nodes: 10 5 15 2 7 12 20

Test Case 2: User Input

Enter nodes: 1 2 3 4 5

EXPERIMENT 5 - OUTPUT

Test Case 1: Execution Result

Tree Height: 3

Test Case 2: Execution Result

Tree Height: 5

EXPERIMENT 6 - INPUT

Test Case 1: User Input

Build tree with: 5 3 8 2 4

Test Case 2: User Input

Build tree with: 1 2 3 4 5

EXPERIMENT 6 - OUTPUT

Test Case 1: Execution Result

Balanced: Yes

Test Case 2: Execution Result

Balanced: No (Skewed)

EXPERIMENT 7 - INPUT

Test Case 1: User Input

Enter BST values: 50 30 70 20 40
Search Key: 40

Test Case 2: User Input

Enter BST values: 50 30 70 20 40
Search Key: 90

EXPERIMENT 7 - OUTPUT

Test Case 1: Execution Result

Result: 40 found in BST.

Test Case 2: Execution Result

Result: 90 not found in BST.

EXPERIMENT 8 - INPUT

Test Case 1: User Input

Current BST: 10 5 15
Insert: 7

Test Case 2: User Input

Current BST: 10 5 15
Insert: 20

EXPERIMENT 8 - OUTPUT

Test Case 1: Execution Result

Updated BST (Inorder): 5 7 10 15

Test Case 2: Execution Result

Updated BST (Inorder): 5 10 15 20

EXPERIMENT 9 - INPUT

Test Case 1: User Input

Current BST: 40 20 60 10 30
Delete: 20

Test Case 2: User Input

Current BST: 40 20 60 10 30
Delete: 40

EXPERIMENT 9 - OUTPUT

Test Case 1: Execution Result

Updated BST (Inorder): 10 30 40 60

Test Case 2: Execution Result

Updated BST (Inorder): 10 20 30 60

EXPERIMENT 10 - INPUT

Test Case 1: User Input

Array: 12 11 13 5 6 7

Test Case 2: User Input

Array: 4 10 3 5 1

EXPERIMENT 10 - OUTPUT

Test Case 1: Execution Result

Max-Heap Sort: 5 6 7 11 12 13

Test Case 2: Execution Result

Max-Heap Sort: 1 3 4 5 10

EXPERIMENT 11 - INPUT

Test Case 1: User Input

Tree nodes: 1 2 3 4 5 6

Test Case 2: User Input

Tree nodes: 10 20 30 40

EXPERIMENT 11 - OUTPUT

Test Case 1: Execution Result

Level-order: 1 2 3 4 5 6

Test Case 2: Execution Result

Level-order: 10 20 30 40

EXPERIMENT 12 - INPUT

Test Case 1: User Input

Enter tree: 10 5 15 2 7 12 20

Test Case 2: User Input

Enter tree: 1 2 3

EXPERIMENT 12 - OUTPUT

Test Case 1: Execution Result

Leaf nodes: 2, 7, 12, 20

Test Case 2: Execution Result

Leaf nodes: 2, 3

EXPERIMENT 13 - INPUT

Test Case 1: User Input

Tree A: 1 2 3 | Tree B: 1 2 3

Test Case 2: User Input

Tree A: 1 2 3 | Tree B: 1 2 4

EXPERIMENT 13 - OUTPUT

Test Case 1: Execution Result

Status: Identical

Test Case 2: Execution Result

Status: Not Identical

EXPERIMENT 14 - INPUT

Test Case 1: User Input

Vertices: 3

Edges: (0,1), (1,2)

Test Case 2: User Input

Vertices: 4

Edges: (0,1), (0,2), (0,3)

EXPERIMENT 14 - OUTPUT

Test Case 1: Execution Result

```
0: 1
1: 0, 2
2: 1
```

Test Case 2: Execution Result

```
0: 1, 2, 3
1: 0
2: 0
3: 0
```

EXPERIMENT 15 - INPUT

Test Case 1: User Input

Edges: 0-1, 0-2, 1-3
Start: 0

Test Case 2: User Input

Edges: A-B, B-C, C-A
Start: A

EXPERIMENT 15 - OUTPUT

Test Case 1: Execution Result

DFS: 0 1 3 2

Test Case 2: Execution Result

DFS: A B C

EXPERIMENT 16 - INPUT

Test Case 1: User Input

Edges: 0-1, 0-2, 1-3
Start: 0

Test Case 2: User Input

Edges: 1-2, 1-3, 2-4
Start: 1

EXPERIMENT 16 - OUTPUT

Test Case 1: Execution Result

BFS: 0 1 2 3

Test Case 2: Execution Result

BFS: 1 2 3 4

EXPERIMENT 17 - INPUT

Test Case 1: User Input

Source: 0, Dest: 3
Path via (0,1), (1,2), (2,3)

Test Case 2: User Input

Source: 0, Dest: 4
Path via (0,1), (2,3)

EXPERIMENT 17 - OUTPUT

Test Case 1: Execution Result

Path Found: Yes

Test Case 2: Execution Result

Path Found: No

EXPERIMENT 18 - INPUT

Test Case 1: User Input

Vertex: 0

Graph: 0-1, 1-2

Test Case 2: User Input

Vertex: 4

Graph: 4-5, 6-7

EXPERIMENT 18 - OUTPUT

Test Case 1: Execution Result

Reachable: 1, 2

Test Case 2: Execution Result

Reachable: 5

EXPERIMENT 19 - INPUT

Test Case 1: User Input

Undirected Edges: (1,2), (2,3), (3,1)

Test Case 2: User Input

Undirected Edges: (1,2), (2,3)

EXPERIMENT 19 - OUTPUT

Test Case 1: Execution Result

Cycle: Yes

Test Case 2: Execution Result

Cycle: No

EXPERIMENT 20 - INPUT

Test Case 1: User Input

Directed Edges: $(0,1)$, $(1,2)$, $(2,0)$

Test Case 2: User Input

Directed Edges: $(0,1)$, $(0,2)$

EXPERIMENT 20 - OUTPUT

Test Case 1: Execution Result

Cycle: Yes

Test Case 2: Execution Result

Cycle: No

EXPERIMENT 21 - INPUT

Test Case 1: User Input

Edges: (A, B) , (B, C) , (A, C)

Test Case 2: User Input

Edges: $(0, 1)$, $(0, 2)$, $(0, 3)$

EXPERIMENT 21 - OUTPUT

Test Case 1: Execution Result

A: 2, B: 2, C: 2

Test Case 2: Execution Result

0: 3, 1: 1, 2: 1, 3: 1

EXPERIMENT 22 - INPUT

Test Case 1: User Input

Nodes 1-5, Edges (1,2), (4,5)

Test Case 2: User Input

Nodes 1-3, Edges (1,2), (2,3)

EXPERIMENT 22 - OUTPUT

Test Case 1: Execution Result

Components: 3

Test Case 2: Execution Result

Components: 1

EXPERIMENT 23 - INPUT

Test Case 1: User Input

Start 0, Weights: (0,1,4), (0,2,1), (2,1,2)

Test Case 2: User Input

Start A, Weights: (A,B,10), (A,C,5), (C,B,2)

EXPERIMENT 23 - OUTPUT

Test Case 1: Execution Result

```
Dist to 1: 3, Dist to 2: 1
```

Test Case 2: Execution Result

```
Dist to B: 7, Dist to C: 5
```

EXPERIMENT 24 - INPUT

Test Case 1: User Input

Adjacency Matrix [0 2 0; 2 0 3; 0 3 0]

Test Case 2: User Input

Adjacency Matrix [0 4 3; 4 0 1; 3 1 0]

EXPERIMENT 24 - OUTPUT

Test Case 1: Execution Result

MST Edges: 0-1 (2), 1-2 (3)

Test Case 2: Execution Result

MST Edges: 1-2 (1), 0-2 (3)

EXPERIMENT 25 - INPUT

Test Case 1: User Input

Edges with Wt: (1,2,4), (2,3,2), (1,3,1)

Test Case 2: User Input

Edges with Wt: (A,B,10), (B,C,5), (A,C,15)

EXPERIMENT 25 - OUTPUT

Test Case 1: Execution Result

Total MST weight: 3

Test Case 2: Execution Result

Total MST weight: 15

EXPERIMENT 26 - INPUT

Test Case 1: User Input

```
Matrix:  
0 3 INF  
2 0 INF  
INF 7 0
```

Test Case 2: User Input

```
Matrix:  
0 5  
INF 0
```

EXPERIMENT 26 - OUTPUT

Test Case 1: Execution Result

```
Result:  
0 3 INF  
2 0 INF  
9 7 0
```

Test Case 2: Execution Result

```
Result:  
0 5  
INF 0
```

EXPERIMENT 27 - INPUT

Test Case 1: User Input

Edges: $(0, 1, 5)$, $(1, 2, -2)$, $(2, 0, 3)$

Test Case 2: User Input

Edges: $(A, B, 1)$, $(B, C, -5)$, $(C, A, 2)$

EXPERIMENT 27 - OUTPUT

Test Case 1: Execution Result

```
Dist: 0:0, 1:5, 2:3
```

Test Case 2: Execution Result

```
Negative Cycle Detected
```

EXPERIMENT 28 - INPUT

Test Case 1: User Input

X: ABCDE, Y: ACE

Test Case 2: User Input

X: AGGTAB, Y: GXTXAYB

EXPERIMENT 28 - OUTPUT

Test Case 1: Execution Result

LCS: ACE, Length: 3

Test Case 2: Execution Result

LCS: GTAB, Length: 4

EXPERIMENT 29 - INPUT

Test Case 1: User Input

Set: {1, 3, 5, 2}, Sum: 8

Test Case 2: User Input

Set: {10, 7, 5, 18}, Sum: 35

EXPERIMENT 29 - OUTPUT

Test Case 1: Execution Result

Solution: {1, 5, 2}

Test Case 2: Execution Result

Solution: {10, 7, 18}

EXPERIMENT 30 - INPUT

Test Case 1: User Input

Operation: Insert 50, 30; Search 30

Test Case 2: User Input

Operation: Insert 10; Search 20

EXPERIMENT 30 - OUTPUT

Test Case 1: Execution Result

Inserted. 30 found.

Test Case 2: Execution Result

Inserted. 20 not found.